Supplementary Information

for: Participants or repeated trials? Equal-budget learning surfaces and an evaluated sequential acquisition rule in three gait datasets

[Authors: to be completed by the authors]

S1. Execution log, verification and amendments

Stage	Model fits	Policy-internal fits	Inner-tuning fits	Refits	Exclusion-control fits	Independent verifier
FPI mechanical pilot	50	-	-	-	50	pass (listed checks)
FPI learning surface (A1)	5,400	-	-	-	5,400	pass (listed checks)
FPI inner-tuning sensitivity (A2)	-	-	5,400	600	600	pass (listed checks)
GaitRec mechanical pilot	50	-	-	-	50	pass (listed checks)
GaitRec learning surface (A3)	4,500	-	-	-	4,500	pass (listed checks)
FPI + GaitRec sequential rule (A4)	5,250	243,161	-	-	5,250	pass (listed checks)
StepUP pilot (A5)	50	-	-	-	50	pass (listed checks)
StepUP surface, sex (A5)	6,000	-	-	-	6,000	pass (listed checks)
StepUP surface, BMI (A5)	6,000	-	-	-	6,000	pass (listed checks)
StepUP sensitivity, sex (A5)	-	-	5,400	600	-	pass (listed checks)
StepUP sequential rule, sex (A5)	2,700	166,050	-	-	2,700	pass (listed checks)
StepUP sequential rule, BMI (A5)	2,700	166,050	-	-	2,700	pass (listed checks)
StepUP small-budget rule, sex (A5.2)	2,700	74,250	-	-	2,700	pass (listed checks)
StepUP small-budget rule, BMI (A5.2)	2,700	74,250	-	-	2,700	pass (listed checks)

Counts are model fits whose results enter the tables (final models and, where stated, policy-internal, inner-tuning and refit fits) and the exclusion-control duplicates. Across all stages there were 38,100 final model fits, 723,761 policy-internal fits, 10,800 inner-tuning fits, 1,200 refits and 38,700 exclusion-control fits. The StepUP sensitivity stage had no exclusion-control duplicates. "Pass (listed checks)" means that the independent verifier passed the checks described in Section 2.9 and no more.

Amendments, each written before any performance result of its stage was read:

- A4.1: the internal balanced accuracy of the sequential rule is computed over the classes present in the inner-validation fold, and an undefined internal estimate defaults to recruiting (the first pilot had silently resolved undefined comparisons to "trials").
- A4.2: the first full policy grid stopped when a class-blind acquired set lacked a class; models then assign probability zero to the absent class and are counted.
- A5.1: with 150 people and five folds each training set has 120 people, so StepUP class allocation was computed per fold from that fold's class counts.
- A5.2.1 and A5.2.2: in the small-budget follow-up the largest fixed designs acquire only 5 to 15 people, which cannot support 12 principal components or, sometimes, two classes; the number of components is min(12, n − 1) and a single-class set yields a constant predictor.

Verifier corrections: the FPI learning-surface verifier first failed because scikit-learn stores a scale of 1 for zero-variance columns, and it was corrected; the policy verifier was corrected for the order of budgets and for ties at the precision stored in the decision log (294 ties accepted); two smaller defects (a pandas attribute lookup and a cell absent from the pilot grid) were corrected. No result file was changed by any verifier correction. Superseded or aborted runs were preserved with their records: three superseded

policy pilots, the first policy grid (aborted on a missing class), one StepUP pilot (aborted on class allocation) and three small-budget attempts (aborted). The first learning-surface pilot attempt was aborted by the exclusion test, because a median function that skips missing values had selected trials by their missing status; its partial output was deleted before the rule of keeping failed runs was adopted, and the defect and its correction are recorded in the pilot report.

S2. Protocol hashes

Protocol file	Field	SHA-256 (recorded before execution)
frozen_protocol_A1.json	protocol_md_sha256	d98ee18951e3d0fe3e3f62f09d459211890570b31d757e9283035826b6f5a623
frozen_protocol_A1.json	addendum_sha256	1e43e2c0da3c6b2265a063b997dc947503dd27f52b3a611c486f7455b7ceebae
frozen_protocol_A1.json	grid_script_sha256	6f25db2781578141dcb57404d3cf7c459a2c587c114fb5a3a04a05bc4e682475
frozen_protocol_A2.json	addendum_A2_sha256	c9a788f058d517861be434608cac3845958151a9c8731c5126541221b4f60119
frozen_protocol_A2.json	script_sha256	0b71b1ca619b03dcf36138471a8dc974d49f8cddacf9b47d64aad94c2b4a1891
frozen_protocol_A3.json	addendum_A3_sha256	b86692d409e0f3ac8a8849f38fd2e5e9acc5734d049a268f082dfce5f7df0185
frozen_protocol_A3.json	script_sha256	8b6e8b474a990a1152025392b784d8894581ec6cea9493133e0c02241a52e37e
frozen_protocol_A4.json	addendum_A4_sha256	f29b8d14bf9d00e0a91d70dd89c75b9c518b248e11b47a9b3b8cb768aa57776a
frozen_protocol_A4.json	script_sha256	8784077192577e679e1d3ec975e2a273772f79daddb74e6731da090903d15ffc
frozen_protocol_A5.json	addendum_A5_sha256	1669d9910f99c6576a0e731ebb7df46f7560776e57da753a212f94f2a9999985
frozen_protocol_A5.json	script_sha256	0a00bd52172b41a00d100ba6f5b16a11af1bc54348517c283e7daabe3735451e
frozen_protocol_A5_2.json	addendum_A5_2_sha256	40b1240b5eba65d56939ea2a20b62ce8cc99cc10444ce7bc0e31e5f50c1c8c21
frozen_protocol_A5_2.json	script_sha256	49c40e9cacf3e748850b228a231506912c191443c4aa1922fc7e29f68a869111
frozen_protocol.json	protocol_md_sha256	d98ee18951e3d0fe3e3f62f09d459211890570b31d757e9283035826b6f5a623

S3. FPI learning surface (balanced accuracy [95% CI])

People (n)	k = 1	k = 3	k = 5
21	0.610 [0.547, 0.670]	0.621 [0.555, 0.684]	0.622 [0.556, 0.684]
28	0.636 [0.564, 0.703]	0.648 [0.574, 0.717]	0.648 [0.574, 0.718]
35	0.654 [0.577, 0.727]	0.659 [0.577, 0.733]	0.664 [0.581, 0.739]
42	0.671 [0.587, 0.749]	0.679 [0.592, 0.758]	0.678 [0.591, 0.758]
49	0.682 [0.593, 0.764]	0.691 [0.601, 0.776]	0.696 [0.605, 0.780]
56	0.691 [0.596, 0.780]	0.699 [0.601, 0.791]	0.722 [0.624, 0.815]

S4. GaitRec learning surface (balanced accuracy [95% CI])

People (n)	k = 1	k = 3	k = 5
300	0.404 [0.385, 0.423]	0.408 [0.389, 0.427]	0.408 [0.389, 0.428]
500	0.414 [0.393, 0.435]	0.416 [0.395, 0.437]	0.418 [0.397, 0.440]
600	0.417 [0.396, 0.439]	0.421 [0.399, 0.443]	0.422 [0.399, 0.444]
800	0.422 [0.400, 0.445]	0.427 [0.404, 0.451]	0.427 [0.404, 0.451]

People (n)	k = 1	k = 3	k = 5
1000	0.427 [0.404, 0.451]	0.430 [0.406, 0.455]	0.431 [0.407, 0.457]

S5. StepUP sex learning surface (balanced accuracy [95% CI])

People (n)	k = 1	k = 3	k = 5	k = 10	k = 20
30	0.530 [0.507, 0.553]	0.543 [0.516, 0.568]	0.553 [0.524, 0.580]	0.556 [0.525, 0.585]	0.557 [0.526, 0.587]
60	0.526 [0.494, 0.556]	0.549 [0.513, 0.584]	0.550 [0.511, 0.587]	0.555 [0.513, 0.595]	0.561 [0.519, 0.602]
90	0.532 [0.494, 0.568]	0.553 [0.510, 0.594]	0.562 [0.516, 0.606]	0.567 [0.518, 0.614]	0.572 [0.520, 0.621]
120	0.535 [0.492, 0.575]	0.553 [0.502, 0.601]	0.562 [0.508, 0.613]	0.580 [0.521, 0.636]	0.606 [0.539, 0.668]

S6. StepUP BMI learning surface (balanced accuracy [95% CI])

People (n)	k = 1	k = 3	k = 5	k = 10	k = 20
30	0.598 [0.562, 0.633]	0.619 [0.580, 0.657]	0.618 [0.578, 0.656]	0.616 [0.577, 0.656]	0.618 [0.578, 0.658]
60	0.633 [0.585, 0.679]	0.653 [0.602, 0.703]	0.660 [0.610, 0.710]	0.653 [0.600, 0.705]	0.654 [0.600, 0.707]
90	0.654 [0.600, 0.707]	0.665 [0.607, 0.721]	0.671 [0.614, 0.727]	0.669 [0.611, 0.728]	0.664 [0.602, 0.724]
120	0.667 [0.610, 0.722]	0.674 [0.612, 0.733]	0.679 [0.618, 0.739]	0.671 [0.606, 0.735]	0.658 [0.587, 0.727]

S7. Gain from three to five trials per person

Task	People (n)	Gain from 3 to 5 trials, points [95% CI]
FPI posture classes (70 men)	21	+0.1 [-0.8, +0.9]
FPI posture classes (70 men)	28	+0.0 [-0.7, +0.8]
FPI posture classes (70 men)	35	+0.5 [-0.4, +1.4]
FPI posture classes (70 men)	42	-0.1 [-1.0, +0.9]
FPI posture classes (70 men)	49	+0.4 [-0.6, +1.6]
FPI posture classes (70 men)	56	+2.3 [+0.2, +4.9]
GaitRec diagnosis classes (1,301 patients)	300	+0.0 [-0.2, +0.2]
GaitRec diagnosis classes (1,301 patients)	500	+0.3 [+0.1, +0.4]
GaitRec diagnosis classes (1,301 patients)	600	+0.1 [-0.1, +0.3]
GaitRec diagnosis classes (1,301 patients)	800	+0.0 [-0.2, +0.2]
GaitRec diagnosis classes (1,301 patients)	1000	+0.2 [-0.1, +0.4]
StepUP sex (150)	30	+1.0 [-0.1, +2.1]
StepUP sex (150)	60	+0.1 [-1.0, +1.2]
StepUP sex (150)	90	+0.9 [-0.3, +2.0]
StepUP sex (150)	120	+0.9 [-0.3, +2.0]
StepUP BMI ≥ 25 (150)	30	-0.1 [-1.1, +0.8]
StepUP BMI ≥ 25 (150)	60	+0.7 [-0.2, +1.6]
StepUP BMI ≥ 25 (150)	90	+0.6 [-0.3, +1.6]
StepUP BMI ≥ 25 (150)	120	+0.6 [-0.6, +1.7]

S8. FPI contrast variability by repeat

Dataset	Contrast	Mean (points)	Within-repeat SD over 20 draws (points; repeats 1/2/3)	SD of the three repeat means (points)	Pairs favouring A per repeat
FPI	35x3 minus 21x5	+3.7	5.2 / 6.0 / 4.3	0.55	14 / 15 / 16
FPI	56x3 minus 35x5	+3.5	3.8 / 3.8 / 5.3	0.35	16 / 17 / 14

S9. Regularisation chosen in the inner-tuning sensitivity analysis

Dataset	Cell	Inner fits selecting C = 0.1	C = 1	C = 10	Acquired sets
FPI	21x5	474 (79%)	104	22	600
FPI	35x3	498 (83%)	98	4	600
StepUP sex	60x10	299 (100%)	1	0	300
StepUP sex	120x5	298 (99%)	1	1	300

S10. Counterpart on the assessment side (FPI, earlier analysis of the same cohort)

The main analyses vary the training data. An earlier analysis of the FPI cohort varied the number of trials available at assessment while holding the model fixed (a model trained with five trials per person, applied to every subset of k of the five recorded trials of held-out people, with the median of the retained trials as person input; probabilities were never pooled across trials). Balanced accuracy increased from one to five trials, and the class assigned to a person depended on which k trials had been recorded in a large share of assessments; at five trials this share is zero by construction because only one subset exists.

Trials available at assessment (k)	Median-pooled BA [95% CI]	Person-repeat assessments whose class depends on which k trials were recorded
1	0.680 [0.596, 0.760]	39.5%
2	0.692 [0.602, 0.779]	28.6%
3	0.706 [0.616, 0.792]	22.9%
4	0.710 [0.616, 0.800]	12.4%
5	0.722 [0.626, 0.813]	0.0%

With the fixed model, five trials exceeded one trial by 4.3 points (95% CI 0.6 to 8.1), and three trials were 1.6 points below five (−3.6 to 0.6). These intervals are conditional on the fitted models and do not include re-fitting. The comparison shows that the weak value of additional trials at training reported here does not imply that trials are unimportant at assessment.

S11. Data provenance

FPI inputs were rebuilt from the released raw file (SHA-256 6fa44121…9374) so that every descriptor depends on one trial only. Earlier feature tables that contained subject-centred features and preprocessing shared across splits were not used. GaitRec and StepUP inputs were taken from the released files; for StepUP, all 300 source files were verified against the official checksum file (the published checksums of the arrays refer to the uncompressed array payload). Detailed provenance is archived with the study materials.

S12. Literature search

Searches were run in October 2026 in OpenAlex (full-text search, for example "gait classification number of trials per participant number of participants" and "learning curve number of subjects per subject gait OR plantar pressure OR ground reaction force", restricted to 2018 or 2020 onwards), in Crossref and on the web, together with the reference list of Seibel and colleagues and the list of works citing that paper (one citing work in OpenAlex, on heart-rate variability and EEG in mental health). Scopus and Web of Science were not available to us. No retrieved gait, plantar-pressure or ground-reaction-force study varied training participants and trials per participant jointly under participant-disjoint testing. Because the searches were targeted and not systematic, the absence of such a study is stated as a search result, not as proof that none exists.